THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 5: **Describing Motion Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The ratio of the speeds of two bodies of the same mass dropped from different heights 2 m and 8 m just before impact is: 1

(a) 1 : 2 **(b)** 1 : 4 **(c)** 1 : 1 **(d)** 2 : 1

2. Assertion (A): A straight line on a v-t graph inclined to the time axis shows uniform acceleration. Reason (R): The slope of the line is constant. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. A particle moving with uniform acceleration covers 20 m in the first 2 s and 60 m in the next 2 s. Its acceleration is: 1

(a) 20 m/s squared **(b)** 10 m/s squared **(c)** 5 m/s squared **(d)** 2.5 m/s squared

4. The v-t graph of a body is a triangle with base 10 s and height 20 m/s. The distance covered is: 1

(a) 10 m **(b)** 200 m **(c)** 50 m **(d)** 100 m

5. Assertion (A): A car's speedometer measures its average velocity. Reason (R): The speedometer shows the instantaneous speed. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. The displacement of a body is given by s = 5t + 2t squared (SI). Its acceleration is: 1

(a) 2 m/s squared **(b)** 5 m/s squared **(c)** 4 m/s squared **(d)** 9 m/s squared

7. A car starting from rest reaches 25 m/s in 10 s with uniform acceleration and then moves at that speed for 20 s. The total distance is: 1

(a) 625 m **(b)** 750 m **(c)** 125 m **(d)** 500 m

8. Assertion (A): A body can have a velocity directed east and acceleration directed west. Reason (R): The acceleration is opposite to the velocity when the body slows down. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. The displacement of a body is s = 5t + 2t squared (SI units). Its initial velocity is: 1

(a) 9 m/s **(b)** 4 m/s **(c)** 5 m/s **(d)** 2 m/s

10. A lift going up moves with an acceleration a for t1, uniform speed for t2 and retards for t3. The v-t graph is a: 1

(a) triangle **(b)** rectangle **(c)** trapezium **(d)** circle

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A stone is thrown vertically up at 15 m/s from a 20 m-high cliff. Find the velocity when it hits the ground (g = 10). 2

12. Why is the distance covered in successive seconds by a freely falling body increasing? 2

13. Two balls A and B are dropped from heights 5 m and 20 m. Find the ratio of the times and the ratio of the final speeds. 2

14. Show that the distance covered in the last second is the same for the upward and the downward trips for a stone thrown upward in the 1st and last seconds. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The velocity of a body increases from 2 m/s to 14 m/s in 6 s. Find a, s and the velocity at the midpoint of the time. 3

16. Explain why the time taken to fall from the second half of a height is shorter than for the first half using h = 100 m. 3

17. Explain why the speed at the lowest point is the same as the speed at the start if a ball is thrown up and returns to the same level, ignoring air resistance. 3

OR

Compare the motion of two bodies whose s-t graphs are straight lines through the origin with slopes 2 and 5. Find the speeds and the distance between them after 10 s. 3

18. A bike accelerates from 0 to 54 km/h in 5 s. Find the acceleration and the distance. If the acceleration is then halved, find the time to reach 72 km/h. 3

OR

A car is moving at 20 m/s. The driver sees a child 40 m away and brakes with 5 m/s squared after 0.4 s of reaction time. Find the distance and judge if the child is in danger. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The motion of a particle is given by x = 3t squared - 2t + 1 (SI). Find its displacement in the first 3 s, the velocity at t = 3 s, and the average velocity and acceleration. 5

OR

A body is thrown vertically upward from the top of a 40 m high tower with a speed of 20 m/s. Find the maximum height above the ground, the time to reach the ground and the speed with which it hits the ground (g = 10). 5

20. A cricket ball hit vertically upward reaches a maximum height of 45 m. Find the speed of the hit, the time of flight and the velocity 2 s after the hit (g = 10). 5

OR

Discuss the role of reaction time and braking distance in road safety with the equation d = vt + v squared/2a, calculating the total stopping distance at 20 m/s and 30 m/s with t = 0.5 s and a = 5 m/s squared. 5

END OF QUESTION PAPER — Total: 40 Marks